Instructions for oral presentation of Home assignment 2, Task 1

An effective oral presentation is an integral part of this course, this time with the aim of giving you experience conveying the results of a technical investigation to a non-technical audience in a way that they can understand. Whether you choose to stay in academia or pursue a career in industry, the ability to communicate clearly is of paramount importance.

You have 6 minutes to present to the class with roughly 2 additional minutes for questions.

You should make (good) slides that will be distributed to the class. Your goal should be to convey to your audience a clear understanding of your research question, along with a basic understanding of your analysis and findings. You should not tell us everything that you did, or show a bunch of analyses that you eventually scrapped (though sometimes it can be useful to see one solution that doesn't work). After hearing your talk, each person in the audience should be able to answer:

(a) What was your project about?
(b) Which analyses did you run?
(c) What did you find?

There are many sources of advice for how to make a good presentation, but an excellent place to start is: http://techspeaking.denison.edu/

Watch the videos on this site to identify some common mistakes. Here is some general advice:

- Budget your time. You only have 6 minutes. If your talk runs far too short, or too long (we'll cut after 6 minutes), it makes you seem unprepared. Rehearse your talk ahead of time (with your group) several times in order to get a better feel for your timing.
- Don't write too much on each slide. You don't want people to have to read your slides, because if the audience is reading your slides, then they aren't listening to you. You want your slides to provide visual cues to the points that you are making – not substitute for your spoken words. Concentrate on graphical displays and bullet-pointed lists of ideas.
- Tell a story – not necessarily the whole story. It is unrealistic to expect that you can tell your audience everything that you know about your topic in 6 minutes. You should strive to convey the big ideas in a clear fashion, but not dwell on the details. Your talk will be successful if your audience is able to walk away with an understanding of what your research question was, how you addressed it, and what the implications of your findings are.

Present the results of the statistical analysis with a target audience of non-statisticians in mind. It usually helps to imagine that e.g. your parents are in the audience (or any other family member that hasn't studied statistics).

Start your presentation with a summary of the data, describe the variables and which kinds of relationships that have been analyzed.

When presenting tables and/or graphs, go through them in detail! For this reason, keep them as simple as possible and leave out unnecessary details.

Show the results based on conclusions, leave out details such as p-values. Such details could be stored in a back-up slide if the audience should have any questions about them.

Words/phrases/items to avoid in your presentation:	What to do instead:
• Model	Describe investigated relationships
• Dependent/independent	use "outcome" or "response", and "explanatory"
• Metric/nonmetric	Numerical/categorical
• Coefficients	Describe effects
• Estimate	
• Fit	
• Significance level	
• Equations, especially with unknowns (e.g. β)	

Words/phrases that can be used but need to be explained:

- Statistically significant
- Significant effect/relationship

Remember that non-statisticians trust you!
Live up to that! Take charge and let the audience feel that you are confident in what you do.

You will receive individual feedback on your presentations.